Enhancing Retrieval-Augmented Generation. Generalizing Contrastive In-Context Learning

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Abstract

The generation of information through Large Language Models (LLMs) is becoming increasingly central to today's technological landscape, with widespread adoption across industries and among individual users. However, because LLMs rely solely on static training data, they are prone to producing outdated or inaccurate information—a phenomenon known as hallucination. Retrieval-Augmented Generation (RAG) addresses this limitation by incorporating external context into model queries, allowing for more accurate and up-to-date responses.

Recent research has introduced Contrastive In-Context Learning (CICL) as an enhancement to RAG. This approach involves including, within the prompt, a question similar to the input paired with a correct answer, alongside another (possibly identical) question paired with an incorrect answer. By presenting both accurate and inaccurate information, CICL helps the model better distinguish between the two during generation.

The objective of this study is to develop an automated method for generating the question–answer pairs required by CICL from arbitrary data sources, thereby eliminating the need for manually curated datasets. The proposed techniques are empirically validated, and the complete codebase is made publicly available to support reproducibility and further research. A.4

1. Introduction

This work builds upon the study *Enhancing Retrieval-Augmented Generation: A Study of Best Practices* [4], which systematically evaluates strategies to improve the performance of Retrieval-Augmented Generation (RAG) systems. In that study, the authors examine both established and novel techniques within a unified experimental framework, identifying the most effective methods through rigorous comparative analysis. Among the novel contributions, **Contrastive In-Context Learning** (CICL) emerges as the most promising.

CICL augments RAG by enriching the query context with a semantically similar question, accompanied by both a correct and an incorrect answer. This contrastive setup guides the language model toward the desired answer format and facilitates more precise retrieval of relevant information through analogy with prior examples. As a result, CICL significantly improves the quality and relevance of generated responses.

However, the original CICL implementation relies on manually curated and annotated datasets to provide the contrastive question–answer pairs. While this yields strong results in controlled experiments, the dependency on manual annotation limits the scalability and applicability of the method in real-world scenarios.

To overcome this limitation, the present work proposes automated methods for generating the required question–answer pairs. Specifically, two novel approaches are explored: **document-based context extraction** and **automatic Q&A generation**.

The *document-based context extraction* method leverages text fragments as contrastive examples: a *positive* fragment is selected as the one most relevant to the query, while a *negative* fragment is chosen as the k-th most relevant result, where k is a manually defined parameter.

The *automatic Q&A generation* method uses a large language model (LLM) to produce both correct and incorrect question–answer pairs from a given data source. Each element of the source is provided to the LLM as context, along with a prompt instructing the model to generate questions and their corresponding answers.

This report presents the design and implementation details of these two methods, as well as the experimental results obtained by integrating them into the original CICL framework.

2. Architecture

This section presents the architecture of the components used to evaluate the proposed methods. The implementation is based on the framework developed in the study *Enhancing Retrieval-Augmented Generation: A Study of Best Practices*, with the necessary modifications to accommodate the new configurations. Consequently, all other elements remain consistent with those described in the original work.

2.1. Retrieval Mechanism

The retrieval module in our system follows the same approach as described in the original study. FAISS [1] is employed due to its computational efficiency, ease of implementation, and strong performance in large-scale similarity searches within high-dimensional vector spaces.

The documents in the knowledge base are first segmented into chunks $C = \{c_1, c_2, \ldots, c_n\}$. A pre-trained Sentence Transformer encoder [7] is then used to generate embeddings $E = \{e_1, e_2, \ldots, e_n\}$ from these chunks.

An `IndexBuilder` class indexes these embeddings to enable efficient retrieval. Given a query embedding q_{emb}—produced by the same encoder—the system retrieves the top-k most similar chunks according to the inner product similarity:

$$\text{Sim}(q_{\text{emb}}, e_i) = q_{\text{emb}}^{\top} e_i$$

The mechanism ensures that the most semantically relevant content is identified and used as context in subsequent processing stages.

This retrieval system is utilized in both proposed generalization methods. In the Document-Based Context Extraction approach, it retrieves the required k fragments. Although only the first and the k-th fragments are used—while the intermediate fragments are disregarded—the extraction process returns exactly k elements due to the implementation details of the retrieval tool.

In the case of the Automatic Q&A Generation method, the retrieval mechanism selects the question and its corresponding correct answer that best match the query, followed by retrieving a question and its associated incorrect answer.

2.2. Implementation Details

For vector indexing and similarity search, FAISS is employed due to its efficiency and scalability in handling large-scale, high-dimensional vector spaces, as stated in the previous subsection. Sentence embeddings are generated using the pre-trained Sentence Transformer model `all-MiniLM-L6-v2`, which enables effective semantic comparison between text fragments.

Text generation tasks—including question and answer generation in the Automatic Q&A Generation method, as well as answer generation within the Retrieval-Augmented Generation (RAG) framework—are performed using models from the Mistral family [3]. Specifically, the `Instruct7B` model serves as the large language model (LLM). Although not the largest or most powerful available model, `Instruct7B` achieves strong results while maintaining a reasonable computational cost.

The knowledge base consists of Level 3 Wikipedia Vital Articles, which are used for both the Document-Based Context Extraction and the Automatic Q&A Generation approaches. This dataset not only provides a rich and diverse information source suitable for our experimental setup but also matches the dataset used in the original paper, thereby enabling more meaningful and direct comparisons.

3. Experimental Setup

This section presents the details of the experimental setup, including the evaluation datasets and the evaluation metrics. To ensure that the results are as comparable as possible, the same datasets and metrics as those used in the original paper have been employed.

3.1. Evaluation Datasets

The performance of the RAG variants is evaluated using two publicly available datasets: *TruthfulQA* [5] and *MMLU* [2].

These datasets were chosen to represent distinct contexts in which a RAG system may be deployed. *TruthfulQA* focuses on general commonsense knowledge, whereas *MMLU* requires more specialized and precise domain expertise. Employing both datasets enables a comprehensive assessment across a broad range of scenarios.

TruthfulQA contains 817 questions spanning 38 categories (e.g., health, law, politics) and is specifically designed to evaluate the truthfulness of large language models by probing common misconceptions. Each sample consists of a question, the best answer, a set of correct answers, and a set of incorrect answers. The questions and answers generated by the *Automatic Q&A Generation* method adhere to the same structure as this dataset, ensuring full compatibility with the evaluation framework.

MMLU evaluates model performance in educational and professional contexts through multiple-choice questions covering 57 subjects. To balance topic coverage with computational and time constraints, the first 32 examples from each subject are selected, resulting in a total of 1,824 samples for evaluation.

3.2. Evaluation Metrics

A comprehensive evaluation of generative performance is conducted using the following metrics:

ROUGE [4]: A widely used family of metrics for assessing the quality of generated text by measuring its lexical overlap with reference texts. ROUGE-1 F1, ROUGE-2 F1, and ROUGE-L F1 scores are employed to evaluate unigram overlap, bigram overlap, and the longest common subsequence, respectively.

Embedding Cosine Similarity: This metric measures the cosine similarity between embeddings of the generated and reference texts, both encoded using a pre-trained Sentence Transformer model [8]. It captures semantic similarity beyond simple lexical overlap.

MAUVE [6]: A metric designed for evaluating open-ended text generation by comparing the distribution of model-generated text to that of human-written text via divergence frontiers. Texts are embedded using a Sentence Transformer, and MAUVE computes the similarity between their embedding-based distributions. Because MAUVE relies on distribution estimation, its results may be unstable for single or very small sample sizes. To ensure stable and meaningful scores, it is computed over the entire evaluation dataset.

The MAUVE values obtained in this study differ significantly from those reported in the original work. This discrepancy is attributed to the metric's sensitivity to the execution environment, as reproducing the exact computational setup used in the original paper was not possible.

4. Document-Based Context Extraction

The first generalization method proposed in this work is *Document-Based Context Extraction*. This approach uses different text excerpts as context, where the most similar fragment to the query within the knowledge base is selected as the *positive* context, and the k-th most similar fragment is selected as the *negative* context.

The procedure is as follows: the elements stored in the knowledge base are divided into chunks, which serve as the candidate fragments for this method. These fragments are then compared with the query using an indexing mechanism, allowing the retrieval of the k most similar fragments. The tool used for producing the indices and computing similarity scores are described in the *Retrieval Module* subsection of the *Architecture* section 2.1.

The parameter k determines which fragment is selected as the negative context in terms of similarity ranking. A higher value of k increases the semantic discrepancy between the positive and negative contexts. In this study, several values of k were tested to evaluate the impact of similarity differences and to identify the most effective configuration. The values evaluated were $k \in \{2, 4, 8, 16, 32\}$.

Examples of contexts generated with different k values can be seen in the *Appendix* A.1. As k increases, the discrepancy between the fragments becomes more pronounced, although the overall topic often remains similar even for $k = 32$. This is due to the nature of the dataset, which consists of long-form articles; when split into chunks, the 32-nd most similar fragment may still originate from the same document. This behavior is, in fact, desirable, since both the positive and negative contexts should be related to the query, with the distinction between them lying in subtle differences.

Experimental results indicate that the best performance was achieved with $k = 2$, as shown in Table 1. This configuration yields the highest performance across most fields, notably excelling in the TruthfulQA dataset. The

	TruthfulQA					MMLU				
	R1	**R2**	**RL**	**ECS**	**MVE**	**R1**	**R2**	**RL**	**ECS**	**MVE**
DK2	**25.62**	**12.09**	**22.70**	**55.89**	43.02	10.22	1.95	8.73	**29.40**	55.61
DK4	25.32	11.75	22.43	55.49	**44.80**	10.36	1.92	8.84	29.15	57.55
DK8	25.25	11.75	22.32	55.28	44.60	**10.41**	**2.04**	**8.90**	29.22	**66.04**
DK16	**25.62**	11.98	22.54	55.37	41.33	10.10	1.80	8.64	29.23	58.21
DK32	25.49	11.92	22.50	55.71	37.52	10.13	1.88	8.69	29.18	61.66

Table 1: Performance metrics for different values of k in the Document-Based Context Extraction approach.

second-best configuration is $k = 8$, which performs best on the MMLU dataset. However, the performance differences across all tested values of k are not significant.

The superior performance of $k = 2$ is noteworthy because it suggests that the positive and negative contexts are derived from the two fragments most similar to the query. Consequently, it becomes ambiguous whether the observed performance gain is truly due to the exploitation of contrastive context or simply the presence of highly relevant fragments.

Taken together with the comparison between the performance of *Document-Based Context Extraction* and the baseline, these findings suggest that this generalization method is not particularly effective. Therefore, alternative approaches merit exploration. The detailed quantitative results are presented later in the *Results* section 6.

5. Automatic Q&A Generation

The second generalization method proposed in this work aims to generate questions along with their corresponding correct and incorrect answers from a given data source. This enables the creation of material with the same structure as that used in the original RAG-CICL experiments, but derived from any source that meets the system's requirements, rather than being limited to manually crafted datasets with highly specific and narrowly defined content.

The process operates as follows: given a data source, a prompt is submitted to a large language model (LLM) to generate a question related to its content, along with answers in the following structure:

- **Best answer**: the most accurate and precise answer to the question.
- **Correct answers**: two additional answers that are correct but less precise than the best answer.
- **Wrong answers**: two plausible but incorrect answers that could reasonably be given by a human respondent.

All generated material is stored in a document, which is then used as a data source. The prompt employed for generating the questions and answers, along with examples of the generated Q&A pairs, is provided in the *Appendix*. A.2

The results obtained with this approach are highly promising, closely matching those presented in the original work using hand-crafted datasets and outperforming the baseline. The observed discrepancies can be attributed to the LLM used in these experiments, which may lack the capability to produce questions and answers of the same quality as those authored by humans. Furthermore, a relatively small model with only 7 billion parameters was employed for this study. Nevertheless, this method shows strong potential and warrants further investigation, particularly in experiments involving larger and more capable models, as well as different application contexts.

6. Results

This section presents the results of the study, comparing the two proposed generalization approaches with the basic RAG system, as well as with the original implementation described in the reference paper. The latter used the same question–answer datasets (TruthfulQA and MMLU) as the source for the Contrastive In-Context Learning method.

As shown in the results, the original configuration (ICLD+) achieves the best overall performance. This approach uses questions and answers from the same dataset as the evaluation benchmark. Although exact answers to the target question are excluded during the retrieval process, the remaining questions and answers are of high quality, as the datasets are generated under human supervision.

	TruthfulQA					MMLU				
	R1	**R2**	**RL**	**ECS**	**MVE**	**R1**	**R2**	**RL**	**ECS**	**MVE**
Base	25.77	11.96	22.53	55.88	**54.43**	10.41	1.89	8.91	29.42	**65.07**
ICLD+	**27.11**	**14.48**	**24.40**	54.72	35.53	**12.21**	**3.08**	**10.52**	**33.65**	60.71
Q&AG	25.91	12.21	22.67	**56.09**	37.43	11.09	2.45	9.50	30.98	52.80
DK2	25.62	12.09	22.70	55.89	43.02	10.22	1.95	8.73	29.40	55.61

Table 2: Performance comparison of different methods on TruthfulQA and MMLU datasets.

The second-best method is one of the proposals introduced in this work: automatic Q&A generation (Q&AG). Results show that this method outperforms the base RAG configuration and comes closest to the performance of the original Contrastive In-Context Learning approach. The performance gap is likely due to two factors: first, the Q&A pairs were generated using a relatively lightweight model of only 7 billion parameters (as described in the methodology section), and second, the Wikipedia-based data source may not align as closely with the evaluation questions, which can lead to less optimal generated pairs.

Looking at the base configuration (Base) and the best-performing Document-Based Context Extraction setup (K = 2), their results are very similar. This suggests that the document-based CICL approach is not particularly effective. As discussed in its dedicated section, this highlights the importance of the context format: a question–answer configuration appears to be significantly more effective than providing longer text fragments.

Taking all these factors into account, we conclude that automatic Q&A generation is a promising avenue for further research. It shows the potential to approach the performance of the original method while being fully generalizable — enabling Q&A generation from any data source — and still has room for improvement, either by refining the generation prompts or by employing more powerful models.

7 Future Work

For future work, several directions could be pursued. As previously discussed, it would be of particular interest to evaluate the automatic Q&A generation method using a larger-scale LLM. The model used in this study has a relatively modest number of parameters, and deeper architectures tend to produce higher-quality generations, which could lead to more accurate and relevant question–answer pairs.

Another promising avenue would be to experiment with different data sources for generating Q&A pairs. In this study, only level-3 Wikipedia articles were used, without exploring alternative sources. Evaluating performance across multiple datasets — beyond the two considered here (MMLU and TruthfulQA) — would help determine whether the method is truly generalizable and to what extent its behavior varies depending on the domain and data characteristics.

Finally, a fine-tuned generation model trained on the TruthfulQA dataset could be explored. The effectiveness of the TruthfulQA format has already been demonstrated in the original paper. Fine-tuning a model on this dataset could yield generations that closely match the style and quality of the original, human-curated Q&A pairs, potentially improving the performance of the method even further.

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A Appendix

The appendix presents examples of the contexts generated using both proposed approaches: Document-Based Context Extraction and Q&A Generation. It also includes the prompt employed for generating question–answer pairs in the generative approach. Finally, the code repository is provided, containing the framework used for the experiments as well as the script for Q&A generation.

A.1 Document-Based Context Generation Examples

The following are examples of contexts retrieved for different values of K, illustrating how the negative contexts diverge progressively as K increases.

- **Query:** *Why do veins appear blue?*

- **Positive Context:** *Veins close to the surface of the skin appear blue for a variety of reasons. However, the factors that contribute to this alteration of color perception are related to the light-scattering properties of the skin.*

  The positive context remains the same across all K values, since the fragment with the highest similarity to the query is unaffected by the number of retrieved contexts.

- $K = 2$**, Negative Context:** *This is because the spectrum of light absorbed by hemoglobin differs between the oxygenated and deoxygenated states. Blood in carbon monoxide poisoning is bright red, because carbon monoxide causes the formation of carboxyhemoglobin.*

- $K = 4$**, Negative Context:** *Effect colors: the blue of the sky (Rayleigh scattering, caused by structures much smaller than the wavelength of light, in this case, air molecules), the luster of opals, and the blue of human irises.*

- $K = 8$**, Negative Context:** *The human venous system develops mainly from the vitelline veins, the umbilical veins, and the cardinal veins, all of which empty into the sinus venosus. About 98.5% of the oxygen in a sample of arterial blood in a healthy human breathing air at sea-level pressure…*

- $K = 16$**, Negative Context:** *Many crustaceans and a few chelicerates and tracheates use respiratory pigments to assist oxygen transport. The most common respiratory pigment in arthropods is copper-based hemocyanin; this is used by many crustaceans and a few centipedes. Some crustaceans and insects use iron-based hemoglobin, the respiratory pigment used by vertebrates.*

- $K = 32$**, Negative Context:** *The spine at the back of the skeleton contains the flexible vertebral column which surrounds the spinal cord, a collection of nerve fibres connecting the brain to the rest of the body. Nerves connect the spinal cord and brain to the rest of the body.*

As shown in these examples, the higher the value of K, the less related the negative context becomes to the original question, although it retains some thematic connection to provide meaningful contrast. Only excerpts of the full contexts are presented here, as the complete passages were too lengthy to include in their entirety.

A.2 Automatic Q&A Generation Examples

The following examples illustrate questions and answers generated using the Automatic Q&A Generation method, with Level 3 Wikipedia articles as the data source. The format of the generated results follows the structure of the TruthfulQA dataset.

- **Question 1:** Who was Hammurabi and what is he best known for?

- **Best Answer:** Hammurabi was an Amorite king of Babylon (1792–1750 BC), best known for issuing the Code of Hammurabi, a legal code that emphasized physical punishment for offenders.
- **Correct Answers:** [Hammurabi was a king of Babylon (1810–1750 BC) known for issuing the Code of Hammurabi., Hammurabi is best known for his military conquests and the Code of Hammurabi.]
- **Incorrect Answers:** [Hammurabi was a Pharaoh of Egypt., Hammurabi was a Greek philosopher.]

- **Question 2:** Who was Alexander the Great's tutor before his assumption of kingship over Macedon?
  - **Best Answer:** Aristotle
  - **Correct Answers:** [Aristotle, The philosopher Aristotle tutored Alexander before his assumption of kingship over Macedon]
  - **Incorrect Answers:** [Plato, Isocrates]

As shown in these examples, the *Best Answer* field provides a single, most concise and accurate response. The *Correct Answers* field lists two additional valid responses, while the *Incorrect Answers* field contains two plausible but factually incorrect responses, which could realistically be given by a human due to their contextual relevance to the correct answers.

A.3 Prompt Used for the Automatic Q&A Generation

Prompt:

```
Generate a challenging question based on:

"{data}"

Include:
- one best answer,
- two correct paraphrases,
- two plausible wrong answers.

Answers max 15 words, no explanations or markdown
    .
Output ONLY this JSON format:

{
  "question": "...",
  "best_answer": "...",
  "correct_answers": ["...", "..."],
  "incorrect_answers": ["...", "..."]
}

Example:

{
  "question": "What is the capital of Canada?",
  "best_answer": "Ottawa",
  "correct_answers": ["Ottawa", "The capital is
      Ottawa"],
  "incorrect_answers": ["Toronto", "Vancouver"]
}
```

In this prompt, the {data} parameter corresponds to the content of a Level 3 Wikipedia article, which serves as the basis for generating the question and its associated answers. The output is strictly formatted in JSON to facilitate subsequent data processing and integration into the main RAG system script. Finally, the example provided includes an entry from the original TruthfulQA dataset, guiding the model toward producing results aligned with the expected output structure.

A.4 Project Repository

Project Name: RAG_CICL-TUM
Owner: surinyach
Repository: https://github.com/surinyach/RAG_CICL/